

TASK 2: Medium – Matrix Operations

C Programming – Code

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#include <stdio.h>

#define MAX 10

void printMatrix(int a[MAX][MAX], int r, int c) {
    int i, j;
    for (i = 0; i < r; i++) {
        for (j = 0; j < c; j++)
            printf("%d ", a[i][j]);
        printf("\n");
    }
}

int main() {
    int a[MAX][MAX], b[MAX][MAX], sum[MAX][MAX];
    int product[MAX][MAX], transpose[MAX][MAX];
    int r1, c1, r2, c2, i, j, k;

    printf("Enter rows and columns of Matrix A: ");
    scanf("%d %d", &r1, &c1);

    printf("Enter Matrix A elements:\n");
    for (i = 0; i < r1; i++)
        for (j = 0; j < c1; j++)
            scanf("%d", &a[i][j]);

    printf("Enter rows and columns of Matrix B: ");
    scanf("%d %d", &r2, &c2);

    printf("Enter Matrix B elements:\n");
    for (i = 0; i < r2; i++)
        for (j = 0; j < c2; j++)
            scanf("%d", &b[i][j]);

    /* Matrix Addition */
    if (r1 == r2 && c1 == c2) {
        for (i = 0; i < r1; i++)
            for (j = 0; j < c1; j++)
                sum[i][j] = a[i][j] + b[i][j];

        printf("\nMatrix Addition:\n");
        printMatrix(sum, r1, c1);
    } else {
        printf("\nAddition not possible: dimensions must match.\n");
    }

    /* Matrix Multiplication */
    if (c1 == r2) {
        for (i = 0; i < r1; i++)
            for (j = 0; j < c2; j++) {
                product[i][j] = 0;
                for (k = 0; k < c1; k++)
                    product[i][j] += a[i][k] * b[k][j];
            }

        printf("\nMatrix Multiplication:\n");
        printMatrix(product, r1, c2);
    } else {
        printf("\nMultiplication not possible: columns of A must equal rows of B.\n");
    }

    /* Transpose of Matrix A */
    for (i = 0; i < r1; i++)
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        for (j = 0; j < c1; j++)
            transpose[j][i] = a[i][j];

    printf("\nTranspose of Matrix A:\n");
    printMatrix(transpose, c1, r1);

    return 0;
}
```